

SEISMIC ANALYSIS DESIGN OF FRAMES WITH VISCOELASTIC CONNECTIONS

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ABSTRACT: A new viscoelastic-type connection isolator (damper), made of elastomeric material, is presented in this paper. Based on experimental results, a Kelvin-Voigt-type model was developed to simulate the behavior of the connection. By comparing the stress-strain curves of the model with those obtained from the tests, it was found that the durometer of the elastomeric is the predominant material parameter. An analytical method for the analysis of unbraced- and braced-frame structures equipped with those viscoelastic-type connections is presented. The effectiveness of these isolation connections (dampers) is studied numerically by comparing the responses to various dynamic excitations of structures equipped with these dampers with the responses of structures without isolation. The results of these comparisons show that the isolation connections provide significant improvement by reducing the lateral displacement of the structures.

INTRODUCTION

The use of isolation devices to reduce the structural response to earthquake excitation has attracted considerable attention recently. Isolation devices can be used in two ways, as base isolators (Kelly 1982; Mostaghel and Khodaverdian 1987; Kelly and Koh 1990; Ahmadi et al. 1990) and as connection isolators (damper). The connection isolators are aiming at dissipating the earthquake energy by placing the isolation device at selected connections of the structure. Various dissipating devices have been proposed in the past. Among the best known are the friction-type and the viscoelastic-type dissipators.

Shake-table tests have shown that the installation of mechanical-energy-dissipating devices improves earthquake resistance of structures considerably (Baktash et al. 1983; Pall 1986; Baktash and Marsh 1986). Various configurations of structures with energy-dissipating devices in the form of friction braces (Pall and Marsh 1982) or viscoelastic dampers (Mahmoodi 1969, 1974; Gasparini et al. 1980, 1981; Zhang et al. 1989) have been studied in the last two decades. It has been shown that the incorporation of these devices into the structure (especially high-rise buildings) is beneficial, improving response to wind and earthquake (Mahmoodi and Keel 1986; Lin et al. 1991).

This paper describes a new type of energy-dissipating connection, developed and tested recently (Hsu and Fafitis 1990). Based on test results, a Kelvin-Voigt-type model was developed, in which the elastic modulus is expressed as function of the durometer of the elastomeric material, and the viscosity coefficient is related to the durometer hardness and frequency. An analytical method for the dynamic analysis of structures with these energy-dissipating connections is also presented, and some results are evaluated and discussed.

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EXPERIMENTAL DATA

The energy-dissipating connection studied in this paper is shown in Fig. 1, and it consists of two single-tooth devices symmetrically placed. The shear force is transferred through the shear pin, thus the energy-dissipating device is subjected only to axial force. The properties of this device were studied experimentally. A typical specimen is shown in Fig. 2. The variables that were studied in this set of tests were the durometer and thickness of the elastomeric pads and the frequency of the cyclic sinusoidal loading. The strain is defined as the deformation of the pad divided by the thickness of the pad. The stress is the total force divided by the area of the pad. The force causing compression on the bottom pad is defined as positive.

The load function applied in this set of tests was sine wave function with maximum displacement of 30% of the elastomeric pad's thickness. The frequencies of the sine wave were 0.25 Hz, 0.5 Hz, 2 Hz, and 4 Hz. The durometers, D , tested were 40, 60, 80, 90, and 95, thus ranging from soft to hard elastomeric. The thicknesses studied were 9/16, 7/16, and 5/16 in. for the bottom pads, and 9/32, 7/32, and 5/32 in. for the top pads. The test results, time, load, and displacement were recorded using a data-acquisition system at a maximum frequency of 100 points per sec. All these tests were carried out under room temperature, and the effect of the temperature was not included in this study. Some typical response curves are shown in Figs. 3–5.

CONNECTION MODELING

The Kelvin-Voigt model was used to model the behavior of the specimens. The parameters of the model were determined by using least-square method. The Kelvin-Voigt model consists of a spring and dashpot connected in parallel. The equilibrium equation is expressed as

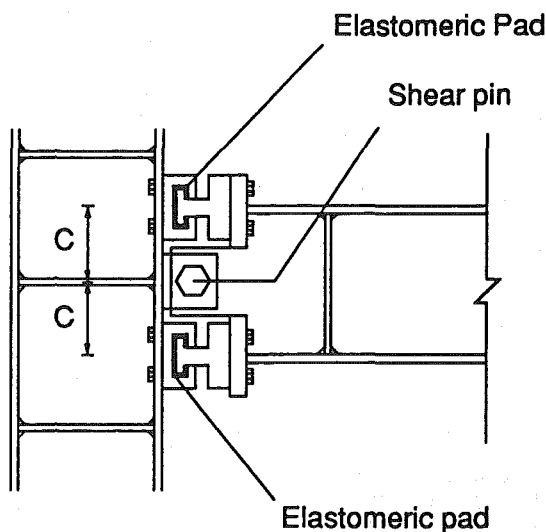


FIG. 1. Beam-Column Connection with Energy-Dissipating Device

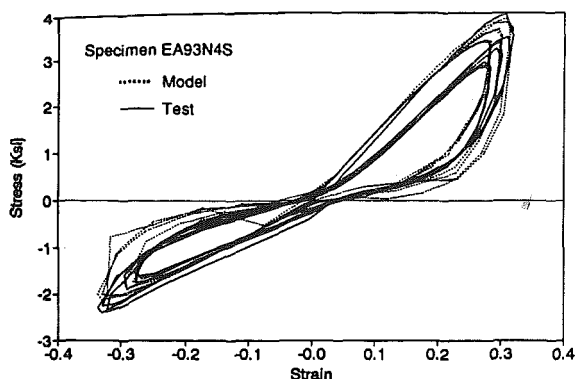


FIG. 3. Experimental and Analytical Stress-Strain Curves with Durometer 95; Thickness of Elastomeric Pads; Top, 9/32 in., Bottom, 9/16 in.; Tested at 4 Hz

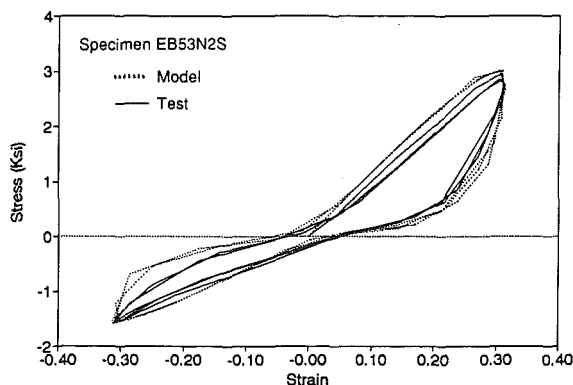


FIG. 4. Experimental and Analytical Stress-Strain Curves with Durometer 90; Thickness of Elastomeric Pads; Top, 5/32 in., Bottom, 5/16 in.; Tested at 2 Hz

The available test results indicate that at constant durometer and frequency, there is a linear relationship of the viscosity coefficient with the strain. Based on this observation, the viscosity coefficient was expressed as

$$\eta_i = A \varepsilon_i \quad \dots \dots \dots (5)$$

where A = a viscosity factor to be determined from experimental data in the sequel.

Once the elastic modulus and the viscosity coefficient are determined, (2) can be used as the stress-strain relationship of the single-tooth connection.

ELASTIC MODULUS AND VISCOSITY COEFFICIENT

The dependence of the elastic modulus and viscosity factor on the durometer was determined from the available experimental data (Hsu and Fafitis 1990). Based on these test data, the elastic modulus was expressed as a third-order polynomial.

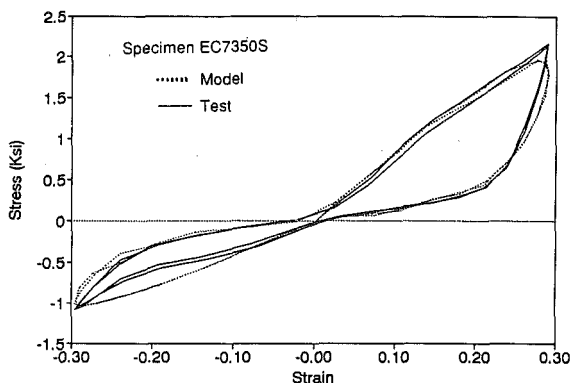


FIG. 5. Experimental and Analytical Stress-Strain Curves with Durometer 80; Thickness of Elastomeric Pads, Top, 7/32 in., Bottom, 7/16 in.; Tested at 0.5 Hz

$$E = -44.504 + 2.44D - 0.0392D^2 + 0.00021D^3, \quad 40 \leq D \leq 95 \quad (6)$$

where E is in ksi. D = the durometer hardness, the measure of the hardness of elastomeric materials. The durometer is scaled from 0 (soft) to 100 (hard). In the present study, durometers from 40 to 95 were used.

The viscosity coefficient was expressed by (5), in which the viscosity factor A was found to vary inversely proportionally to frequency, whereas its dependence on the durometer was expressed by a third-degree polynomial. Thus, by regression analysis, it was found that the viscosity factor A can be expressed as follows.

For loading

$$A = \frac{5.032 - 0.245D + 0.00399D^2 - 0.0000188D^3}{f} \quad (7)$$

for unloading

$$A = \frac{-25.175 + 1.372D - 0.022D^2 + 0.000118D^3}{f} \quad (8)$$

where f = the frequency in Hz and D = the durometer.

Fig. 6 shows the correlations of the elastic modulus and the viscosity factor A with the available experimental data.

GEOMETRIC EFFECT

Because of the unsymmetrical configuration of the device, the compressive and tensile responses of the device are different. It is assumed that the elastic modulus and viscosity coefficient for tension can be estimated by multiplying the elastic modulus and viscosity coefficient for compression by a shape factor. Hence, the elastic modulus and the viscosity factor A for tension become

$$E_c = SE_t \quad (9)$$

$$A_c = S^2A_t \quad (10)$$

where E_c , E_t , A_c , and A_t = elastic modulus and viscosity factor A for

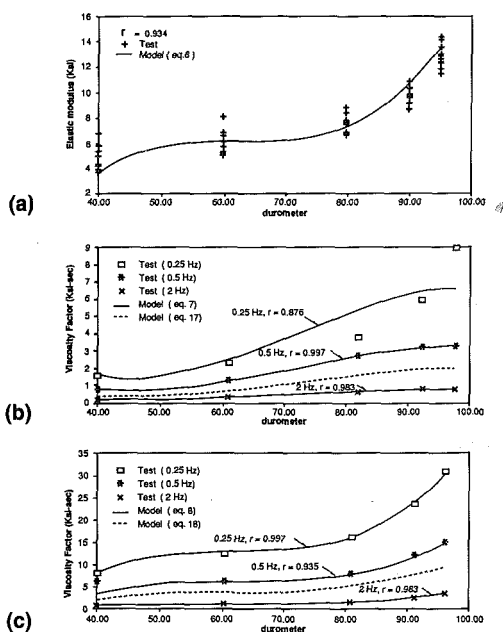


FIG. 6. Correlation between Test Data and Model (r = Correlation Coefficient): (a) Elastic Modulus versus Durometer; (b) Viscosity Factor versus Durometer (Loading); (c) Viscosity Factor versus Durometer (Unloading)

compression and tension, respectively; and S = a shape constant that, for the configuration studied in this work, is assumed to be equal to 1.9.

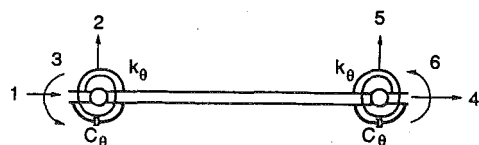
EXPERIMENTAL VERIFICATION

To examine the accuracy of the model developed herein, the stress-strain curve calculated by applying the same input function as in the test is compared with the actual, experimentally obtained stress-strain curve. Figs. 3, 4, and 5 show the comparisons. The agreement between the stress-strain curves predicted by the model and those obtained experimentally indicates that the model approximates the behavior of the single-tooth viscoelastic isolation connection with satisfactory accuracy.

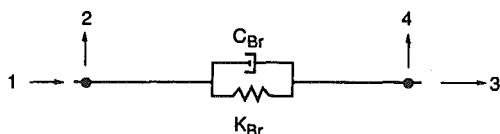
STRUCTURAL MODEL

Unbraced Frame

In an energy-dissipating beam-column connection, the device described earlier is attached symmetrically to the top and bottom flange of the beam and to the column, as shown in Fig. 1. The shear is transferred through a pinned connection. Thus the dissipators are subjected to axial load only, provided that the rotations of the hinge are relatively small. This connection is modeled as shown in Fig. 7(a), where the elastic rheological properties of the beam element are modeled by the twisting springs and dashpots. This modeling is consistent with the traditional concept of plastic hinge. The



(a)



(b)

FIG. 7. Mechanical Modeling: (a) Beam Element with Single-Tooth Connection; (b) Bracing Element with Single-Tooth Connection

frames of Figs. 8(a) and (b) are equipped with these beam-column connections.

Modeling M - θ Relationship of Connection

To establish the moment-rotation relationship for the beam-column connection, the following assumptions are made: The stresses are uniformly distributed over the elastomeric pad, and the stress at the centroid of the pad is taken as the magnitude of this uniform stress. The joint rotation angle, θ , is assumed to be relatively small. The equilibrium equation of the moment, M , about the center of the hinge is expressed as

$$M = \sigma_c a_c C + \sigma_t a_t C \quad (11)$$

where a_c and a_t = the areas of the elastomeric material in compression and tension; σ_c and σ_t = the compression and tension stresses of the elastomeric material; and C = the distance from the center of the hinge to the centerline of each of the two devices, as shown in Fig. 1. Note that compression in each of the two devices is defined as the axial force causing compression of the large (bottom) pad of Fig. 2.

Substituting (1) into (11) gives

$$M = (E_c \epsilon_c + \eta_c \dot{\epsilon}_c) a_c C + (E_t \epsilon_t + \eta_t \dot{\epsilon}_t) a_t C \quad (12)$$

Substituting (5) into (12) gives

$$M = \left[E_c \frac{C\theta}{h_c} + \left(A_c \frac{C\theta}{h_c} \right) \frac{C\dot{\theta}}{h_c} \right] a_c C + \left[E_t \frac{C\theta}{h_t} + \left(A_t \frac{C\theta}{h_t} \right) \frac{C\dot{\theta}}{h_t} \right] a_t C \quad (13a)$$

$$M = \left[\left(\frac{E_c a_c}{h_c} + \frac{E_t a_t}{h_t} \right) C^2 \right] \theta + \left[\left(\frac{A_c a_c}{h_c^2} + \frac{A_t a_t}{h_t^2} \right) \theta C^3 \right] \dot{\theta} \quad (13b)$$

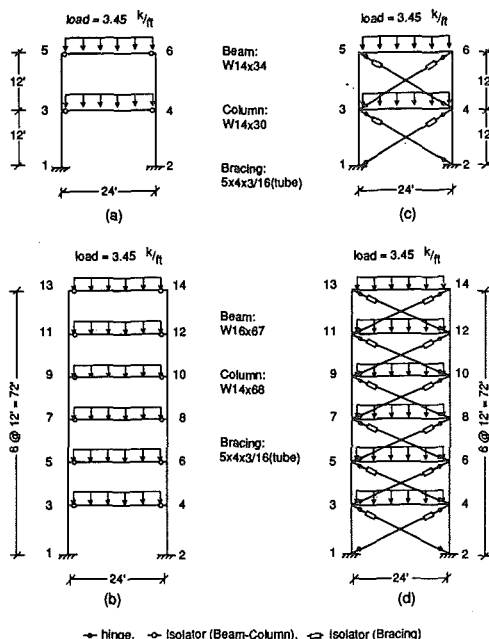


FIG. 8. (a) One-Bay, Two-Story Unbraced Frame; (b) One-Bay, Six-Story Unbraced Frame; (c) One-Bay, Two-Story Braced Frame; (d) One-Bay, Six-Story Braced Frame

$$M = K_0\theta + C_0\dot{\theta} \quad \dots \dots \dots (13c)$$

where E , A , and h = elastic modulus, viscosity factor, and thickness of the elastomeric material. The subscripts c and t indicate the compression and tension, respectively. K_0 and C_0 represent the rotational stiffness and viscous damping of the connection.

Braced Frame

To study the effect of energy dissipators on bracing—and also to compare the performance of frames with dissipative connections, as described in the previous paragraphs, with braced frames—the structures of Figs. 8(c) and (d) were analyzed. These two structures are identical in dimensions and design with those of Figs. 8(a) and (b), except that the energy-dissipating connection were replaced by rigid beam-column connections and cross-bracings with energy dissipators were added, as shown in Figs. 8(c) and (d).

The single-tooth device described earlier is placed at the bracing members. Each diagonal bracing member is assumed to be a one-dimensional element with both ends pinned to beam-column joint of the frame. The degrees of freedom of a bracing element are the two translational displacements at each node, and the member can be subjected to extension or compression forces. Buckling is not considered in this study. The assumed mechanical model for the bracing element is shown in Fig. 7(b).

Modeling F - δ Relationship of Bracing Element

The axial force-displacement relationship of the bracing element consists of the contributions of the elastomeric (which is soft compared with steel),

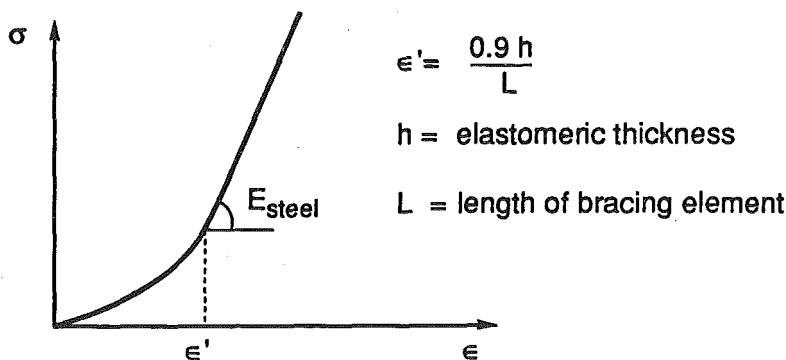


FIG. 9. Typical Stress-Strain Curve of Bracing Element with Single-Tooth Connection

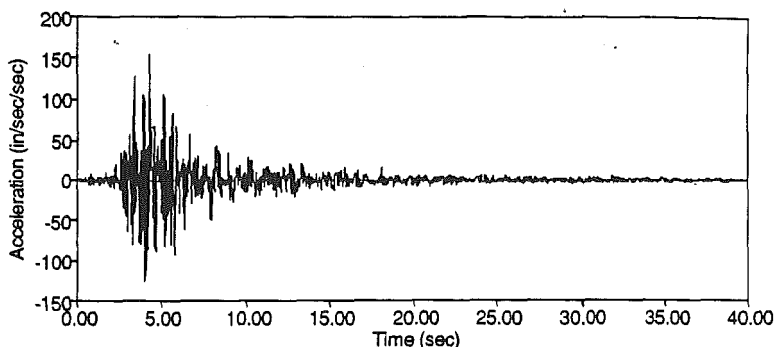


FIG. 10. Acceleration History of Whittier Narrow Earthquake

and the contribution of the steel. A schematic force-displacement relationship is shown in Fig. 9. Note that for strain $\epsilon < \epsilon'$, the strain of the steel is negligible, and one can assume that the deformation of the member is due only to the deformation of the elastomeric. For strains larger than ϵ' , the steel part of the bracing takes over. However, transition from the small elastic modulus of the elastomeric to the large elastic modulus of the steel would cause impact effects, which are to be avoided. It is therefore recommended that the structure be designed in such a way that the strain in the bracing be less than ϵ' . If the bracing strain is kept at these levels, then the axial force in the bracing member is also low, and thus buckling can be ignored. The design of the frames studied in this work was such that these requirements were met, and the strain at the bracings was kept low through the analysis.

Using (1), the force-displacement relationship of the bracing element can be expressed as follows:

For tension

$$F_t = \left(\frac{E_t a_t}{h_t} \right) \delta + \left(\frac{A_t a_t}{h_t^2} \right) \delta = K_t \delta + C_t \delta \quad \dots \dots \dots (14)$$

and for compression

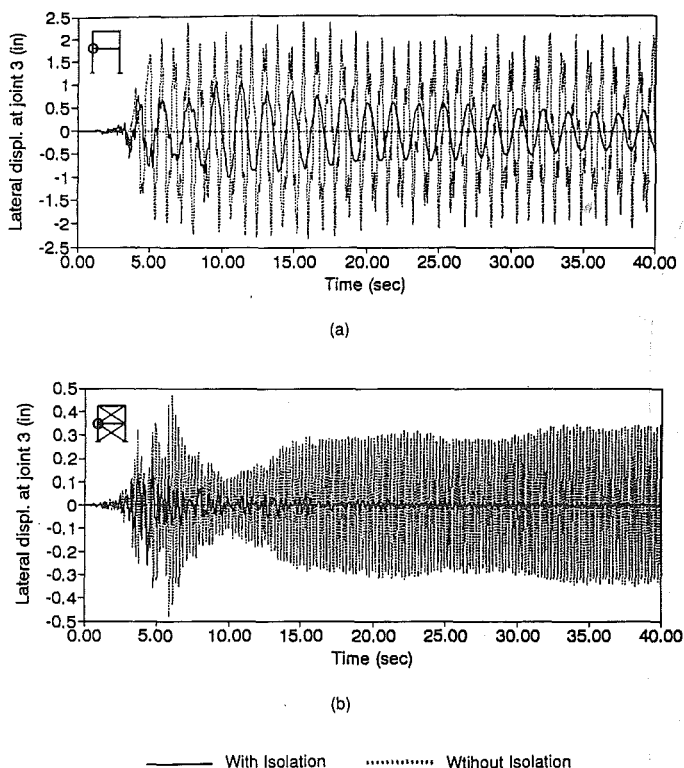


FIG. 11. Response of 2F Frames at Joint 3: (a) Unbraced; (b) Braced

$$F_c = \left(\frac{E_c a_c}{h_c} \right) \delta + \left(\frac{A_c a_c}{h_c^2} \right) \dot{\delta} = K_c \delta + C_c \dot{\delta} \dots \dots \dots (15)$$

where F_t and F_c = axial tensile and compressive forces, respectively; K_c , C_t and C_c = the stiffness and damping matrices of the bracing element for tension and compression, respectively, and δ = the axial deformation of the bracing element.

EQUATION OF MOTION

The differential equation of motion of framed structures equipped with isolation devices can be formulated as follows

$$\ddot{\mathbf{M}}\mathbf{u}(t) + \dot{\mathbf{C}}\mathbf{u}(t) + \mathbf{K}\mathbf{u}(t) = \mathbf{Q}(t) \dots \dots \dots (16)$$

in which $\mathbf{M}$ = the consistent mass matrix; $\mathbf{C}$ = the viscous damping matrix; $\mathbf{K}$ = the stiffness matrix; and $\mathbf{Q}(t)$ and $\mathbf{u}(t)$ = the forcing and the relative displacement vectors, respectively. The dots denote differentiation with respect to time. Since the length of the energy dissipating connection device is small compared with the length of the beam, the mass of the isolation device can be neglected. The damping coefficient of the steel is assumed negligible compared with the damping of the dissipating device. The damp-

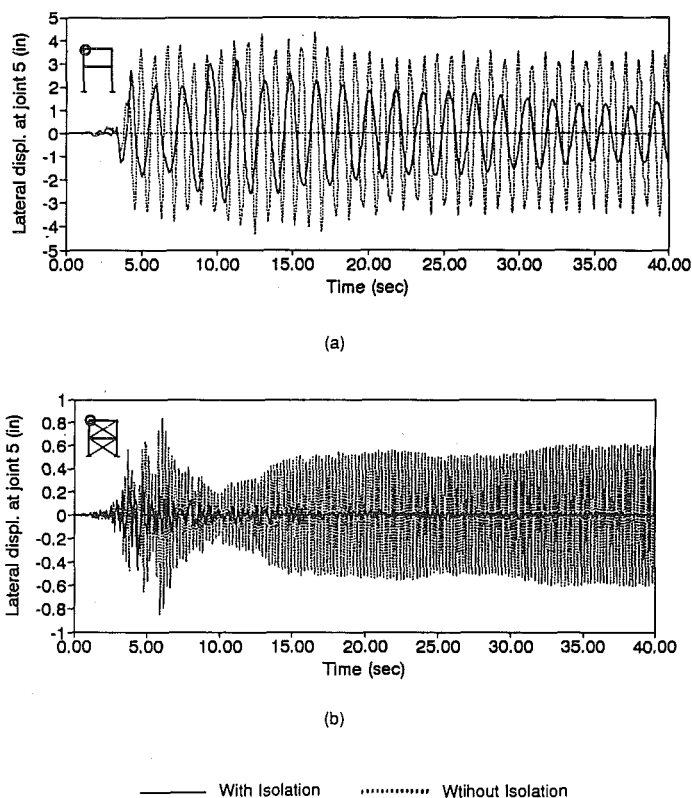


FIG. 12. Response of 2F Frames at Joint 5: (a) Unbraced; (b) Braced

ing coefficient of the connection device or the bracing member are the only damping effects in the structure. The damping matrix consists of the matrices derived from the dissipative devices placed in beam-column connection or the bracing element. For frequencies between 0.5 Hz and 1.5 Hz, it was found that for a given durometer, the viscosity factor A does not vary significantly with frequency. Since this range of frequencies is of interest in earthquake design, it is assumed that within this range the viscosity factor A depends on durometer only, and can be expressed as follows:

For loading

$$A = 5.837 - 0.284D + 0.00463D^2 - 0.0000218D^3 \dots\dots\dots (17)$$

and for unloading

$$A = -29.203 + 1.592D - 0.0255D^2 + 0.000137D^3 \dots\dots\dots (18)$$

In seismic excitation, there is no external load [$Q(t) = 0$], and the loading is due to ground motion. The differential equation of motion [(16)] can be written as follows:

$$\begin{aligned} & (\mathbf{M}_B + \mathbf{M}_{Br})\ddot{\mathbf{u}}(t) + (\mathbf{C}_\theta + \mathbf{C}_{Br})\dot{\mathbf{u}}(t) + (\mathbf{K}_B + \mathbf{K}_\theta + \mathbf{K}_{Br})\mathbf{u}(t) \\ & = -(\mathbf{M}_B + \mathbf{M}_{Br})\ddot{\mathbf{u}}_g(t) \dots\dots\dots (19) \end{aligned}$$

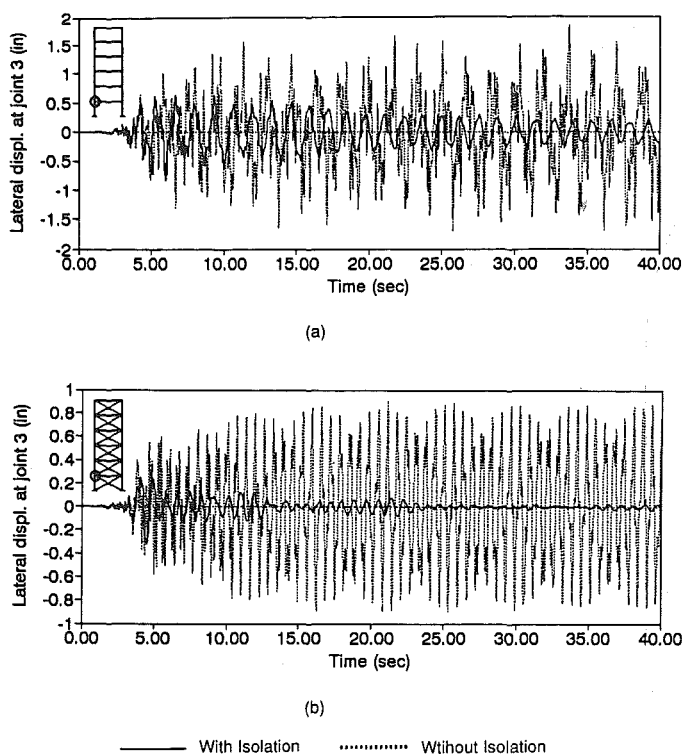


FIG. 13. Response of 6F Frames at Joint 3: (a) Unbraced; (b) Braced

where $\ddot{\mathbf{u}}_g(t)$ = the seismic horizontal acceleration vector; $\mathbf{M}_B$ and $\mathbf{K}_B$ = the consistent mass and stiffness matrices derived from beam and column elements; $\mathbf{M}_{Br}$, $\mathbf{C}_{Br}$, and $\mathbf{K}_{Br}$ = the consistent mass, viscous damping, and stiffness matrices of the bracing elements; and $\mathbf{C}_\theta$ and $\mathbf{K}_\theta$ = the viscous damping and stiffness matrices of the dissipative connection.

APPLICATION

The method described in the previous paragraph is implemented in the analysis of four structures: (a) One-bay, two-story unbraced frame; (b) one-bay, six-story unbraced frame; (c) one-bay, two-story braced frame; and (d) one-bay, six-story braced frame. The fundamental periods of structures (a), (b), (c), and (d), with rigid connections, are 0.89 sec, 1.74 sec, 0.23 sec, and 0.65 sec, respectively. The energy-dissipating devices are placed at all beam-column connections in the unbraced frames, (a) and (b). In frames (c) and (d), the energy-dissipating devices are placed at all bracing elements. The properties and dimensions of these four frames are as shown in Fig. 8, and the durometer of the elastomeric material for all four examples was durometer 90. The thickness of elastomeric pads was 9/16 in. for bottom and 9/32 in. for top.

The time history analysis is carried out using the fourth-order Runge-Kutta method, in which the damping and stiffness matrices to be used in

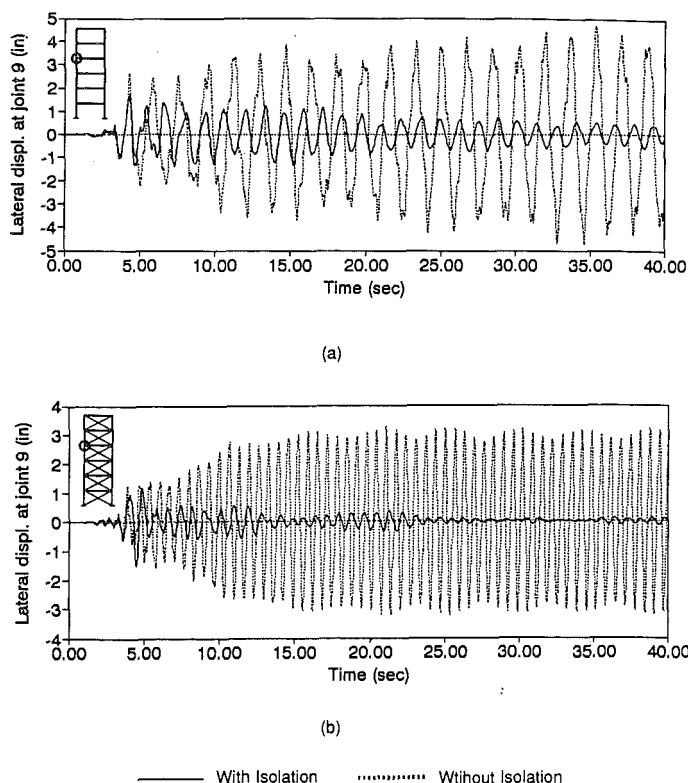


FIG. 14. Response of 6F Frames at Joint 9: (a) Unbraced; (b) Braced

the next step are obtained by updating the elastic modulus and the viscosity factor at each time step. Since the mass matrix is kept constant during the integration procedure, the inverted mass matrix is calculated at the beginning. Therefore, the numerical integration procedure does not require matrix inversion at each time step. To ensure the accuracy of the computation scheme, the numerical integration was repeated by reducing the time step to one-tenth until the difference in the results was insignificant. In this study, a time step in the range 0.003 sec to 0.001 sec was found adequate for the structures shown in Fig. 8. The acceleration of one component of the Whittier Narrow Earthquake (October 1, 1987) is used as the input ground motion for all frames. The acceleration history of this earthquake is shown in Fig. 10.

To demonstrate the effectiveness of the proposed dissipating connection, the response of the frame equipped with the devices is compared with the response of the frame with regular rigid connections. In the analysis of the rigidly connected structures, linear-elastic behavior with zero viscous damping was assumed. The lateral displacements at joints 3 and 5 of frames (a) and (c) are compared in Figs. 11 and 12. The lateral displacements at joints 3, 9, and 13 of frames (b) and (d) are compared in Figs. 13–15.

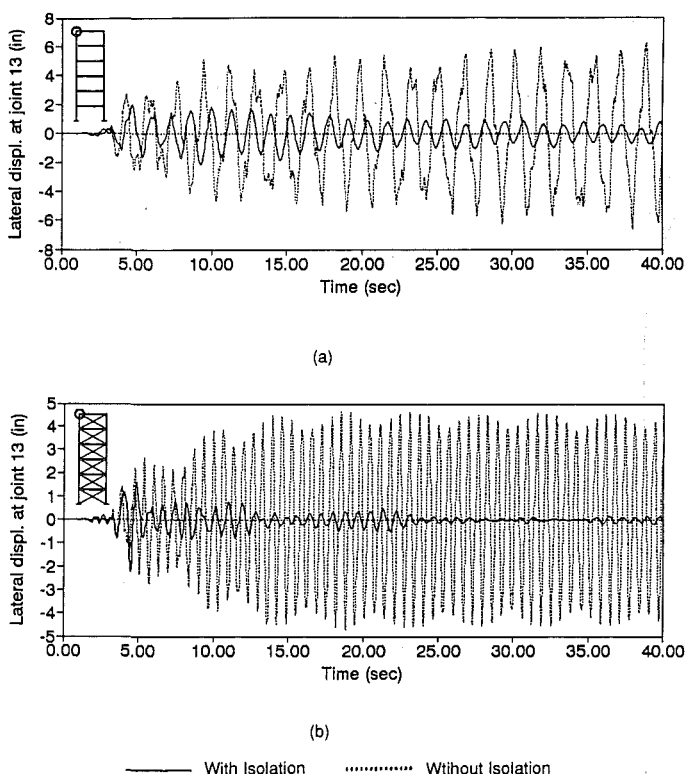


FIG. 15. Response of 6F Frames at Joint 13: (a) Unbraced; (b) Braced

CONCLUSIONS AND DESIGN RECOMMENDATIONS

An analytical model is presented for the dynamic analysis and design of the framed structures with energy-dissipating devices. Based on physical considerations and test data, the moment-rotation relationship for the beam-column connections with dissipating isolator and the force-displacement relationship for bracing elements with dissipating isolators were established. Incorporating the traditional concept of plastic hinge with the moment-rotation relationship of the energy dissipating device, the equations of motion for structures with dissipative connections can be expressed in matrix form. Because of the nonlinearity introduced by the connection, a step-by-step numerical integration procedure was employed to calculate the response of framed structures equipped with this type of connections.

The response to earthquake excitation of braced- and unbraced-frame structures indicate that the lateral displacements of structures equipped with the proposed devices are reduced significantly, compared with structures with rigid connections. It was also found that the response frequencies of the braced and unbraced structures equipped with dissipators were lower than those of structures with rigid connections.

At the present stage, the only method of design that can be suggested is the trial and error, with objectives the control of the lateral static displacement and the dynamic response of the structure. The design parameters

may be: (a) The thickness of the elastomeric pad; (b) the durometer of the elastomeric material; and (c) in the case of framed structures, the distance between the top and bottom flange (C-parameter). The compressive strain of the elastomeric pad should be kept below the 90% limit to avoid impact effects. As a rule of thumb, the expected reduction of the maximum dynamic lateral displacement is 30–60% of the displacement of the rigidly connected structure.

ACKNOWLEDGMENT

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APPENDIX I. CONVERSION TO SI UNITS

<u>To convert</u>	<u>To</u>	<u>Multiply by</u>
in.	mm	25.4
ft	m	0.305
kips	N	4,450
ksi	Mpa	6.9

APPENDIX II. REFERENCES

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